

UK GLP-1 Commercial Opportunity & Disease Burden Analysis

A Power BI Commercial Analytics Project Using NHS Prescribing, Population and QOF Diabetes Data

Executive Summary

This project analyses the commercial dynamics of the GLP-1 diabetes market in England using publicly available NHS data. The analysis combines monthly primary-care prescribing data, GP registered-population data and Quality and Outcomes Framework (QOF) diabetes prevalence data within a Power BI analytical model.

The objective is not only to examine market size and competitive performance, but also to assess whether geographic commercial conclusions change when prescribing activity is adjusted for local disease burden.

The analysis covers January 2025 to June 2026, with year-on-year comparisons based on January to June 2026 versus the corresponding period in 2025.

The core diabetes GLP-1 market recorded approximately 1 million prescription items during January to June 2026, with a Net Ingredient Cost of approximately £87.67 million. Overall prescribing declined by approximately 3.88% year on year.

Semaglutide remained the dominant molecule, representing approximately 75.5% of core diabetes GLP-1 prescribing volume. Dulaglutide remained the second-largest molecule but experienced weaker competitive momentum, while liraglutide demonstrated relatively strong percentage growth from a substantially smaller market base.

A key contribution of the project is the development of a disease-adjusted geographic framework. Standard population-adjusted utilisation was compared with prescribing intensity relative to the recorded QOF diabetes register.

Two analytical indices were constructed:

- Population Market Development Index (MDI)
- Disease-Adjusted Development Index (DADI)

The comparison demonstrates that geographic market development can appear materially different depending on the denominator used. An ICB that appears highly developed when prescribing is divided by general registered population may appear less developed after adjusting for local recorded diabetes burden, and vice versa.

The project therefore highlights an important commercial analytics principle: apparent geographic opportunity should not be assessed using prescribing volume or general population size alone.

ICBs were subsequently segmented using disease-adjusted development and year-on-year growth relative to the England benchmark. This produced four commercial segments:

- Emerging Opportunity
- Expansion Market
- Investigate Barriers
- Established / Mature

The analysis is designed as a commercial decision-support framework rather than a direct sales recommendation. Low utilisation does not automatically represent market whitespace. Differences may reflect local prescribing pathways, disease prevalence, clinical eligibility, formulary policy, access, supply conditions or other structural factors not directly observable in the datasets.

The project demonstrates how pharmaceutical commercial analytics can move beyond descriptive reporting toward a more critical evaluation of market development, competitive dynamics and geographic opportunity.

1. Introduction

The pharmaceutical market increasingly relies on data-driven decision-making to understand competitive position, regional variation and potential commercial opportunity.

In the GLP-1 market, this challenge is particularly important because prescribing patterns are influenced by multiple factors, including disease burden, clinical eligibility, treatment guidelines, product positioning, local prescribing behaviour and the availability of different therapeutic alternatives.

A simple comparison of absolute prescribing volume across regions can therefore be misleading.

A large geographic area may show high prescribing volume simply because it has a larger population. Similarly, a region may appear underdeveloped relative to population size while actually having a relatively low diabetes burden.

For commercial analytics, the denominator used to evaluate market development therefore matters.

This project investigates the GLP-1 diabetes market in England using Power BI and public NHS datasets.

The analysis combines:

- English Prescribing Dataset data
- GP registered-population data
- QOF diabetes prevalence data

The project has three primary analytical objectives.

First, it evaluates market and competitive performance at molecule level.

Second, it examines geographic differences in prescribing intensity across Integrated Care Boards.

Third, it investigates whether geographic commercial interpretation changes when prescribing is adjusted for recorded diabetes burden rather than general population alone.

2. Business Question

The principal business question is:

How is the GLP-1 market evolving across England, which products are gaining competitive position, and which geographic markets show the strongest commercial potential after accounting for local diabetes burden?

This question is supported by five analytical sub-questions:

1. How has the core diabetes GLP-1 market evolved over time?
2. Which molecules are gaining or losing competitive position?
3. How does GLP-1 prescribing intensity vary across Integrated Care Boards?
4. Does the interpretation of geographic market development change when local diabetes burden is incorporated into the denominator?
5. Which geographic areas warrant further commercial investigation based on disease-adjusted development and market growth?

These questions structure the analytical design of the Power BI dashboard.

3. Analytical Scope

3.1 Time Period

The prescribing analysis covers:

January 2025 to June 2026

The principal comparison period is:

January–June 2026 versus January–June 2025

This allows current year-to-date performance to be compared with the equivalent period in the previous year.

Monthly trend analysis retains the complete January 2025 to June 2026 period.

3.2 Geographic Scope

The analysis covers primary-care prescribing across England.

Geographic analysis is conducted at Integrated Care Board level using GP practices as the underlying geographic unit.

Practice-level data was aggregated through a shared practice dimension so prescribing, population and QOF disease-burden information could be evaluated within a consistent geographic model.

3.3 Product Scope

The core analytical market includes the following GLP-1 receptor agonist molecules:

- Semaglutide
- Dulaglutide
- Liraglutide
- Exenatide
- Lixisenatide

However, chemical-level filtering alone was not sufficient.

Several GLP-1 molecules are associated with products used across different treatment pathways, including diabetes and weight management.

For this reason, presentation-level product classification was introduced.

Examples of diabetes-focused brands included:

- Ozempic
- Rybelsus
- Trulicity
- Victoza
- Byetta
- Bydureon

- Lyxumia

Weight-management products were excluded from the core diabetes market where identifiable.

This included products such as:

- Wegovy
- Saxenda

Additional liraglutide and semaglutide presentations were classified using presentation characteristics and known brand mapping.

The resulting analytical framework therefore distinguishes between:

All selected GLP-1 prescribing

and

Core Type 2 Diabetes GLP-1 prescribing

This distinction was necessary because the disease-burden denominator used later in the analysis relates to diabetes rather than the wider obesity market.

4. Data Sources

Three main public datasets were integrated into the analysis.

4.1 NHS English Prescribing Dataset

The primary transactional dataset was the NHS English Prescribing Dataset.

This contains monthly primary-care prescribing activity at GP practice level.

Relevant fields included:

- Practice Code
- Practice Name
- Integrated Care Board
- BNF chemical substance
- BNF presentation
- Presentation description
- Prescription Items
- Quantity
- Net Ingredient Cost

- Actual Cost
- SNOMED Code
- Geographic information

Monthly files covering January 2025 to June 2026 were incorporated.

The prescribing dataset provides the principal measure of market activity.

However, prescription items should not be interpreted as unique patients.

One patient may receive multiple prescription items over time.

4.2 GP Registered Population

Registered population data was obtained from the NHS Patients Registered at a GP Practice dataset.

The June 2026 snapshot was used.

Relevant fields were reduced to:

- Practice Code
- Registered Population

Where necessary, records were aggregated to practice level to ensure a single population value could be related to each GP practice.

The population dataset was used to construct population-adjusted prescribing intensity.

4.3 Quality and Outcomes Framework Diabetes Data

Quality and Outcomes Framework data for the 2025–26 reporting period was used to incorporate local diabetes burden.

The diabetes table provided variables including:

- Practice Code
- Practice Name
- Diabetes Register
- Diabetes Prevalence

The diabetes register represents recorded diabetes burden at GP practice level.

It does not represent the number of patients clinically eligible for GLP-1 treatment.

The register was therefore used only as a disease-burden denominator for comparative analysis.

5. Data Preparation and Transformation

Data preparation was completed primarily in Power Query.

Several transformations were necessary before the datasets could be analysed together.

5.1 Harmonisation of Prescribing Schemas

The NHS prescribing dataset used slightly different column naming conventions across the analysis period.

January and February 2025 used an earlier schema, while later monthly files used a revised format.

For example, equivalent fields appeared under different names across the two structures.

The two groups of files were therefore standardised before being appended.

Key harmonised fields included:

- YearMonth
- RegionName
- RegionCode
- ICBName
- ICBCode
- PracticeName
- PracticeCode
- Postcode
- ChemicalSubstanceCode
- Molecule
- PresentationCode
- PresentationName
- Quantity
- Items
- NIC
- ActualCost
- SNOMEDCode

This ensured that all monthly files could be analysed within one consistent fact table.

5.2 Date Standardisation

Monthly period values were transformed into valid date fields representing the first day of each month.

A separate date dimension was created containing:

- YearMonth
- Year
- MonthNumber
- MonthName
- Quarter
- YearMonthLabel
- YearMonthSort

This allowed correct chronological sorting and time-intelligence calculations.

5.3 Numeric Data Validation

Numeric fields were explicitly checked and standardised.

Particular attention was required for monetary values such as Net Ingredient Cost.

An early validation stage identified a decimal interpretation issue that overstated NIC.

This was corrected by applying the appropriate numeric conversion and locale interpretation in Power Query.

After correction, core diabetes NIC for January to June 2026 was approximately:

£87.67 million

This validation step was important because incorrect locale handling can create significant financial distortions in commercial reporting.

5.4 Product Classification

A dedicated product dimension was constructed using presentation-level information.

The dimension included:

- ChemicalSubstanceCode
- Molecule
- PresentationCode
- PresentationName
- SNOMEDCode
- Brand

- PrimaryMarket
- IncludeDiabetesCore

Products were classified into:

- Type 2 Diabetes
- Weight Management
- Requires Review

A Boolean field, IncludeDiabetesCore, was then used to define the analytical diabetes market.

This approach reduced the risk of treating all prescribing for a molecule as diabetes-related.

5.5 Practice Dimension

A practice dimension was created to provide a common geographic structure.

The most recently observed geographic attributes were retained for each practice.

The dimension contained:

- PracticeCode
- PracticeName
- Postcode
- PCOName
- PCOCode
- ICBName
- ICBCode
- RegionName
- RegionCode

Duplicate practice codes were removed to ensure one row per practice.

This dimension subsequently connected prescribing, registered population and QOF disease data.

6. Data Model

The analytical model was designed around a dimensional structure.

The principal tables were:

- Fact_Prescribing

- Dim_Date
- Dim_Product
- Dim_Practice
- Practice_Population_202606
- QOF_Diabetes_202526

The prescribing fact table contains transactional monthly prescribing activity.

Dimension tables provide time, product and geographic context.

The population and QOF tables provide additional practice-level denominator information.

Core relationships include:

Dim_Date → Fact_Prescribing

using YearMonth.

Dim_Product → Fact_Prescribing

using PresentationCode.

Dim_Practice → Fact_Prescribing

using PracticeCode.

Dim_Practice → Practice_Population_202606

using PracticeCode.

Dim_Practice → QOF_Diabetes_202526

using PracticeCode.

This structure allows multiple forms of analysis without repeatedly merging denominator data directly into the prescribing fact table.

It also supports the separation of market measures from geographic normalisation measures.

7. Analytical Measures

A series of DAX measures was developed to support market, competitive and geographic analysis.

7.1 Core Diabetes Items

Core Diabetes Items represent prescription items associated with products classified as diabetes-related within the product dimension.

Conceptually:

Core Diabetes Items = Total Items filtered to IncludeDiabetesCore = TRUE

This forms the principal market-volume measure.

7.2 Market Share

Molecule-level market share is calculated as:

Molecule Core Diabetes Items / Total Core Diabetes Market Items

This measures the proportion of the defined diabetes GLP-1 market represented by each molecule.

7.3 Year-on-Year Growth

Year-on-year growth is calculated as:

(Current Period Items – Previous Year Items) / Previous Year Items

For the dashboard, January–June 2026 is compared with January–June 2025.

7.4 Market YoY Growth

The same calculation is performed at total category level.

For January–June 2026, the core diabetes market declined by approximately:

3.88% year on year

This value is used as a benchmark when evaluating molecule performance.

7.5 Growth Gap

Growth Gap compares molecule growth with market growth:

Molecule YoY Growth – Market YoY Growth

A positive value indicates that the molecule is outperforming category growth.

A negative value indicates that it is underperforming the market.

The measure is expressed in percentage points.

7.6 Share Change

Share Change measures the movement in market share compared with the prior-year period:

Current Market Share – Previous Year Market Share

It is expressed in percentage points rather than percentage growth.

This distinction is important.

For example, a market share increase from 20% to 22% represents a two percentage-point gain, not 2% growth.

7.7 Growth Contribution

Growth Contribution is calculated as:

Change in Molecule Items / Change in Total Market Items

This measure indicates how much a molecule contributes to overall category change.

Interpretation becomes more complex when total market growth is negative.

If the category contracts, a growing molecule may offset part of that decline rather than positively contribute to category growth in the conventional sense.

For this reason, the dashboard labels the visual as contribution to **net category change**.

8. Executive Market Overview

The Executive Overview dashboard provides a high-level summary of category performance.

The principal KPIs for January–June 2026 were:

- Approximately 1 million core diabetes prescription items
- Approximately -3.88% YoY market growth
- Approximately £87.67 million NIC
- Semaglutide as the leading molecule
- Approximately 75.5% market share for Semaglutide

The overview also includes monthly prescribing trends, market share by molecule, molecule-level growth and contribution to net category change.

8.1 Market Contraction

The core diabetes GLP-1 market declined by approximately 3.88% compared with the corresponding period in 2025.

This indicates that category-level prescribing did not expand during the analysed period.

However, total category contraction does not imply that all molecules declined.

Competitive performance differed substantially across molecules.

8.2 Semaglutide Market Leadership

Semaglutide remained the dominant molecule in the market.

Its approximately 75.5% share means that roughly three quarters of core diabetes GLP-1 prescribing volume was associated with semaglutide within the defined dataset.

As a result, category-level performance is strongly influenced by semaglutide.

Changes in its prescribing volume can therefore have a disproportionate effect on total category trends.

8.3 Competitive Concentration

The market appears highly concentrated around a small number of molecules.

Semaglutide accounts for the majority of prescribing, while dulaglutide represents the principal secondary molecule.

Other therapies account for substantially smaller shares.

This creates an important distinction between:

- percentage growth
- absolute market impact

A small molecule may record high percentage growth while still contributing relatively little to total category volume.

9. Competitive Performance Analysis

The Competitive Performance page extends beyond market share and evaluates competitive momentum.

It incorporates:

- Market Share
- Share Change
- YoY Growth

- Growth Gap
- Growth Contribution
- Current prescribing volume

The central visual is a competitive performance matrix.

The matrix uses:

X-axis: Growth Gap

Y-axis: Share Change

Bubble size: Core Diabetes Items

This allows competitive position to be evaluated along three dimensions simultaneously.

9.1 Semaglutide

Semaglutide combines dominant market share with continued competitive strength.

Its position reflects not only its existing scale but also its ability to maintain or improve share relative to competing molecules.

This distinguishes it from a molecule that is large only because of historical market position.

Semaglutide therefore represents both:

- a market leader
- a key determinant of overall category dynamics

9.2 Dulaglutide

Dulaglutide remains the second-largest molecule within the core market.

However, its competitive momentum is weaker.

Its decline in market share and relative underperformance compared with total market growth indicate that its competitive position has deteriorated during the analysed period.

This does not independently establish the cause.

Possible explanations may include competitive substitution, changing prescribing preferences, availability, clinical guidance or other market factors.

The dataset does not provide sufficient evidence to determine which mechanism is responsible.

9.3 Liraglutide

Liraglutide demonstrates strong percentage growth relative to the market.

However, the commercial interpretation must account for its small starting base.

Large percentage increases from a low baseline can appear dramatic while representing a relatively small change in absolute prescription volume.

For this reason, percentage growth should be interpreted alongside:

- market share
- absolute prescribing volume
- share change
- growth contribution

This prevents overestimating the commercial significance of small-base growth.

9.4 Exenatide and Lixisenatide

Exenatide and lixisenatide represent very small portions of the analysed market.

In the filtered 2026 period, lixisenatide recorded no prescribing activity within the selected primary-care dataset.

This should not be interpreted as evidence that the medicine has been formally discontinued or withdrawn.

The data only supports the narrower conclusion that no prescribing activity was recorded within the defined dataset and period.

10. Geographic Market Development

Absolute prescribing volume alone is insufficient for geographic comparison.

Larger ICBs will generally contain more patients and may therefore generate greater prescribing volume even if relative utilisation is not particularly high.

To address this, prescribing was first normalised by registered population.

11. Population-Adjusted Utilisation

Population-adjusted prescribing intensity is calculated as:

Core Diabetes Items / Registered Population × 1,000

This produces:

Core Diabetes Items per 1,000 registered population

The measure allows more meaningful comparison between differently sized geographic areas.

However, it still assumes that general population provides an appropriate denominator for assessing diabetes-related prescribing.

This assumption is potentially problematic because diabetes burden is not evenly distributed across England.

12. Population Market Development Index

A population-based market development benchmark was therefore constructed.

Population MDI is calculated as:

ICB Core Diabetes Items per 1,000 population / England Core Diabetes Items per 1,000 population × 100

The England benchmark equals 100.

Interpretation is:

- MDI above 100 = utilisation above England average
- MDI equal to 100 = approximately equal to England average
- MDI below 100 = utilisation below England average

Population MDI provides an intuitive geographic benchmark.

However, it remains sensitive to variation in underlying diabetes prevalence.

This motivated the development of a disease-adjusted alternative.

13. Disease-Adjusted Utilisation Rate

The Disease-Adjusted Utilisation Rate, or DAUR, is calculated as:

Core Diabetes Items / Recorded Diabetes Register × 1,000

The purpose is to evaluate prescribing relative to recorded diabetes burden rather than total registered population.

This provides a more disease-specific comparative denominator.

However, DAUR must be interpreted carefully.

It is not a treatment penetration measure.

The numerator contains prescription items rather than unique treated patients.

The denominator represents the QOF diabetes register rather than the number of patients eligible for GLP-1 treatment.

Therefore, DAUR should be interpreted as:

a comparative prescribing-intensity proxy relative to recorded diabetes burden

rather than:

the percentage of diabetes patients receiving GLP-1 treatment

This distinction is central to the analytical integrity of the project.

14. Disease-Adjusted Development Index

The Disease-Adjusted Development Index normalises DAUR against the England benchmark.

It is calculated as:

ICB DAUR / England DAUR × 100

England therefore equals 100.

Interpretation is:

- DADI above 100 = disease-adjusted utilisation above England
- DADI equal to 100 = approximately in line with England
- DADI below 100 = disease-adjusted utilisation below England

DADI provides a comparative geographic development measure that incorporates recorded diabetes burden.

15. Denominator Sensitivity Analysis

One of the central analytical contributions of the project is the comparison between:

- Population MDI
- DADI

This asks whether geographic conclusions change when the denominator changes.

The difference is expressed as:

DADI – Population MDI

A positive gap indicates that a market appears more developed after adjusting for diabetes burden.

A negative gap indicates that a market appears less developed after disease adjustment.

A small gap indicates that both denominators produce broadly similar interpretations.

This analysis demonstrates that commercial conclusions can be sensitive to denominator choice.

15.1 Why the Denominator Matters

Consider two ICBs with identical numbers of GLP-1 prescription items.

If one ICB has substantially higher diabetes prevalence, the same prescribing volume represents lower prescribing intensity relative to local disease burden.

Population-only analysis may fail to detect this.

Conversely, an ICB with relatively low diabetes prevalence could appear only moderately developed on a population basis but relatively highly developed after disease adjustment.

The denominator is therefore not simply a technical calculation choice.

It can materially influence commercial interpretation.

15.2 Commercial Implication

A geographic territory should not automatically be classified as underdeveloped simply because its prescribing volume per general population is low.

Similarly, high population-adjusted prescribing does not automatically mean the market is highly developed relative to disease burden.

Disease prevalence adds context.

This supports a broader principle in pharmaceutical analytics:

commercial opportunity should be assessed relative to the population that is relevant to the disease, not only the population that exists geographically.

16. ICB Growth Analysis

Geographic growth was calculated using core diabetes prescribing among practices with valid geographic coverage.

ICB year-on-year growth compares January–June 2026 with January–June 2025.

The corresponding England geographic benchmark was approximately:

-3.51%

This differs slightly from the overall category figure because geographic measures include only practices eligible for the geographic denominator analysis.

This benchmark is used in segmentation.

Importantly, “high growth” in the segmentation does not necessarily mean positive growth.

An ICB declining by 1% would still be performing above an England benchmark of -3.51%.

Growth is therefore interpreted relatively rather than against zero.

17. Commercial Opportunity Segmentation

ICBs were segmented using two analytical dimensions:

Disease-Adjusted Development Index

and

ICB YoY Growth relative to England

The framework creates four groups.

17.1 Emerging Opportunity

Criteria:

DADI below 100

and

growth equal to or above England

These ICBs remain below the national disease-adjusted development benchmark but are performing better than the overall geographic market.

This may indicate accelerating adoption from a relatively underdeveloped base.

However, this classification should be treated as an analytical flag rather than an automatic investment recommendation.

Further investigation would be required.

17.2 Expansion Market

Criteria:

DADI equal to or above 100

and

growth equal to or above England

These ICBs combine relatively high disease-adjusted utilisation with comparatively strong momentum.

From a commercial perspective, they may represent markets where prescribing is already well developed but still performing strongly.

These areas may warrant continued engagement and monitoring.

17.3 Investigate Barriers

Criteria:

DADI below 100

and

growth below England

These markets display both low relative utilisation and weak growth.

It would be tempting to interpret these areas as commercial whitespace.

However, that conclusion cannot be made from the available data alone.

Low development may reflect:

- local clinical pathways
- formulary restrictions
- prescribing guidance
- supply conditions
- patient eligibility
- demographics
- access limitations
- genuine lower commercial potential

These territories should therefore be investigated before additional commercial resources are allocated.

17.4 Established / Mature

Criteria:

DADI equal to or above 100

and

growth below England

These markets already show relatively strong utilisation but weaker incremental momentum.

This may be consistent with more established or mature prescribing patterns.

Commercial priorities in these areas may differ from those in emerging markets.

For example, competitive defence, retention and share protection may become relatively more important than category expansion.

18. Key Commercial Insights

The integrated analysis produces several commercially relevant findings.

18.1 The Core Diabetes GLP-1 Market Contracted

The category declined by approximately 3.88% year on year during January–June 2026.

This means competitive performance must be interpreted within a declining category context.

A molecule that grows modestly may therefore still materially outperform the market.

18.2 Semaglutide Dominates the Category

Semaglutide accounts for approximately 75.5% of core diabetes GLP-1 prescribing.

This creates high category dependence on one molecule.

Its performance therefore has a major influence on overall market trends.

18.3 Competitive Performance Is Divergent

The market is not moving uniformly.

Semaglutide maintains a strong competitive position.

Dulaglutide remains commercially significant but shows weaker momentum.

Liraglutide demonstrates stronger relative growth but from a much smaller base.

This reinforces the need to analyse both percentage growth and absolute scale.

18.4 Geography Matters

Disease-adjusted utilisation differs substantially across ICBs.

The market therefore cannot be interpreted as geographically homogeneous.

Regional prescribing intensity varies even after accounting for recorded diabetes burden.

18.5 Denominator Choice Changes Interpretation

Population MDI and DADI do not always identify the same ICBs as relatively high or low development markets.

This finding is particularly important from an analytical perspective.

Commercial segmentation can become misleading when denominator choice is not critically evaluated.

19. Commercial Recommendations

The following recommendations are based on the analytical patterns observed.

They should be interpreted as areas for further commercial investigation rather than direct sales instructions.

19.1 Protect Established High-Value Markets

ICBs with high DADI and above-market growth represent relatively developed markets that continue to perform strongly.

These areas should be monitored closely for:

- changes in competitive share
- changes in prescribing momentum
- emerging competitive threats
- evidence of category saturation
- changes in local pathways

Maintaining competitive position may be as important as finding new growth.

19.2 Investigate Emerging Markets

Low-DADI markets growing faster than England may warrant deeper investigation.

These areas may represent developing markets where prescribing adoption is increasing from a lower base.

Commercial analysis could examine:

- which molecules are driving growth
- which practices are contributing most
- whether growth is broad-based or concentrated

- whether local formulary conditions have changed
- whether population or prevalence dynamics explain the movement

19.3 Diagnose White Space Before Investing

Low utilisation should not automatically be treated as opportunity.

An ICB with low DADI and weak growth may have structural reasons for lower prescribing.

Before increasing commercial attention, additional evidence should be collected regarding:

- clinical pathways
- formulary position
- treatment eligibility
- socioeconomic characteristics
- patient demographics
- disease prevalence
- healthcare access
- supply conditions

This reduces the risk of confusing low utilisation with addressable commercial whitespace.

19.4 Monitor Competitive Concentration

Semaglutide's dominance means the category is highly dependent on one molecule.

Competitive analysis should therefore distinguish between:

- category growth
- molecule growth
- share redistribution
- absolute volume contribution

A molecule can grow rapidly in percentage terms while remaining commercially small.

Conversely, a modest change in a dominant molecule may have much greater market impact.

20. Limitations

The analysis has several important limitations.

20.1 Prescription Items Are Not Unique Patients

The prescribing dataset measures prescription items.

A patient may receive multiple items during the analysis period.

Therefore, prescription volume cannot be interpreted as patient count.

20.2 NIC Is Not Manufacturer Revenue

Net Ingredient Cost is a prescribing cost measure.

It should not be treated as direct manufacturer revenue, net sales or profit.

The measure is used only as an indicator of prescribing-associated financial scale.

20.3 Diabetes Register Is Not the Eligible Population

The QOF diabetes register reflects recorded diabetes prevalence.

Not all registered diabetes patients are clinically eligible for GLP-1 therapy.

Therefore, the denominator does not represent the true addressable treatment population.

20.4 DAUR Is Not Treatment Penetration

DAUR uses prescription items in the numerator and diabetes-register patients in the denominator.

Because the numerator is not unique treated patients, the measure cannot be interpreted as the percentage of patients receiving GLP-1 therapy.

It is a comparative utilisation proxy only.

20.5 Patient-Level Switching Cannot Be Observed

The data is aggregated.

If one molecule declines while another grows, this may be consistent with substitution.

However, the analysis cannot demonstrate that individual patients switched between therapies.

Patient-level longitudinal data would be required.

20.6 Indication Identification Is Imperfect

Presentation-level classification improves the distinction between diabetes and weight-management prescribing.

However, aggregate prescribing records do not always provide definitive patient-level indication information.

Some ambiguity therefore remains.

20.7 Temporal Alignment Is Imperfect

Prescribing data is monthly.

QOF prevalence is annual.

Registered population is based on a June 2026 snapshot.

The datasets therefore do not measure all variables at exactly the same point in time.

The denominator should be considered contextual rather than perfectly contemporaneous.

20.8 Local Market Drivers Are Not Directly Observed

The project does not directly measure:

- local formulary decisions
- clinician preferences
- medicine supply constraints
- treatment guidelines
- local commissioning policy
- patient adherence
- prescribing restrictions
- promotional activity

These factors may help explain observed geographic variation.

21. Future Development

The analytical framework could be extended in several directions.

21.1 Obesity Prevalence

Incorporating obesity prevalence could help analyse the broader GLP-1 market and improve interpretation where molecules have both diabetes and weight-management indications.

21.2 Demographic Structure

Age distribution could provide additional explanatory power because diabetes burden and treatment eligibility vary by age.

21.3 Socioeconomic Deprivation

Linking practice or ICB data with deprivation indices could help investigate whether utilisation differs systematically across socioeconomic groups.

21.4 Local Formulary Policy

Incorporating local formulary or prescribing-guideline data could help explain why some ICBs show persistently low or high utilisation.

21.5 Indication-Specific Prescribing

More granular indication-level data would reduce uncertainty created by products used across multiple treatment pathways.

21.6 Patient-Level Longitudinal Data

Where legally and appropriately available, patient-level longitudinal data could support analysis of:

- therapy initiation
- persistence
- switching
- discontinuation
- treatment sequencing

This would significantly extend the analytical capability beyond aggregate prescribing data.

21.7 Wider Incretin Market

The project currently focuses on the selected core GLP-1 receptor agonist market.

Future work could expand the competitive definition to include newer dual GIP/GLP-1 therapies and assess how the broader incretin market changes competitive dynamics.

21.8 Market Concentration Measures

Formal concentration measures such as the Herfindahl-Hirschman Index could be incorporated.

This would provide a quantitative measure of market concentration rather than relying only on leading-market-share metrics.

22. Dashboard Architecture

The final Power BI solution contains four principal pages.

22.1 Executive Overview

Purpose:

Provide senior-level visibility of category performance.

Key outputs include:

- Core Diabetes Items
- Market YoY Growth
- NIC
- Leading Molecule
- Leading Molecule Share
- Monthly prescribing trend
- Market share by molecule
- YoY growth
- Contribution to net category change

22.2 Competitive Performance

Purpose:

Evaluate molecule-level competitive momentum.

Key outputs include:

- Leading Molecule
- Leading Share
- Market Growth
- Fastest-Growing Molecule
- Competitive Performance Matrix
- Market Share
- Share Change
- YoY Growth
- Growth Gap
- Competitive Interpretation
- Share Change by Molecule
- Growth Gap by Molecule

22.3 Geographic Opportunity

Purpose:

Evaluate regional market development after accounting for population and disease burden.

Key outputs include:

- Highest DADI ICBs
- Lowest DADI ICBs

- ICB Commercial Opportunity Matrix
- Commercial Opportunity Segmentation
- Population MDI
- DADI
- ICB Growth
- Denominator Sensitivity

22.4 Commercial Insights, Recommendations & Limitations

Purpose:

Translate analytical outputs into concise commercial interpretation.

The page summarises:

- key market insights
- implications for commercial prioritisation
- areas requiring further investigation
- methodological limitations
- future analytical opportunities

23. Skills Demonstrated

This project demonstrates a combination of business intelligence, data engineering, data modelling and commercial analytics skills.

Technical skills include:

- Power BI
- DAX
- Power Query
- Data transformation
- Data harmonisation
- Dimensional modelling
- Time intelligence
- KPI development
- Dashboard design
- Multi-source data integration

Analytical skills include:

- Market share analysis

- Competitive performance analysis
- Year-on-year growth
- Growth benchmarking
- Share momentum analysis
- Contribution analysis
- Market development indexing
- Geographic segmentation
- Disease-adjusted utilisation
- Denominator sensitivity analysis

Commercial skills include:

- Market interpretation
- Opportunity screening
- Competitive positioning
- Geographic prioritisation
- Critical assessment of proxy measures
- Translation of analytical outputs into business recommendations

24. Conclusion

This project demonstrates how publicly available NHS datasets can be integrated into a commercially relevant pharmaceutical analytics framework.

The analysis goes beyond descriptive prescribing reporting by examining competitive performance, relative market momentum, geographic development and the effect of disease burden on market interpretation.

The principal finding is that market development depends not only on prescribing volume but also on the denominator against which that prescribing is assessed.

Population-adjusted utilisation provides useful geographic context, but it can overlook differences in underlying disease burden.

By incorporating QOF diabetes data, the Disease-Adjusted Development Index provides an additional perspective on relative market development.

The comparison between Population MDI and DADI demonstrates that geographic commercial conclusions can change when disease burden is considered.

This has practical implications for market prioritisation.

Low prescribing does not necessarily mean untapped opportunity, while high prescribing does not necessarily mean unusually high development relative to local need.

The competitive analysis also demonstrates the importance of combining scale and momentum.

Semaglutide dominates the market and remains the principal commercial driver, while smaller molecules may display large percentage movements without comparable absolute impact.

Overall, the project illustrates how pharmaceutical commercial analytics can become more decision-relevant when market share, growth, geography, disease burden and analytical limitations are considered together.

The resulting framework should therefore be understood as a tool for prioritising further investigation rather than producing deterministic commercial conclusions.

References / Data Sources

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Appendix A – Core Measures

Core Diabetes Items

Total prescription items associated with presentations classified as core diabetes GLP-1 products.

Core Diabetes Market Share

Core Diabetes Items for the selected molecule divided by total Core Diabetes Market Items.

Core Diabetes YoY Growth

(Current Core Diabetes Items – Prior-Year Core Diabetes Items) / Prior-Year Core Diabetes Items.

Core Diabetes Market YoY Growth

Total market growth compared with the equivalent prior-year period.

Growth Gap

Molecule YoY Growth – Market YoY Growth.

Share Change

Current Market Share – Previous-Year Market Share, expressed in percentage points.

Growth Contribution

Change in Molecule Items / Change in Total Market Items.

Core Diabetes Items per 1,000

Geo Core Diabetes Items / Registered Population × 1,000.

Population MDI

ICB Core Diabetes Items per 1,000 / England Core Diabetes Items per 1,000 × 100.

DAUR

Disease-Adjusted Core Diabetes Items / Diabetes Register × 1,000.

DADI

ICB DAUR / England DAUR × 100.

DADI vs Population MDI Gap

DADI – Population MDI.

Appendix B – Commercial Segmentation Logic

Emerging Opportunity

DADI < 100

and

ICB YoY Growth ≥ England ICB YoY Growth

Expansion Market

DADI ≥ 100

and

ICB YoY Growth $\geq$ England ICB YoY Growth

Investigate Barriers

DADI < 100

and

ICB YoY Growth $<$ England ICB YoY Growth

Established / Mature

DADI ≥ 100

and

ICB YoY Growth $<$ England ICB YoY Growth

Appendix C – Important Interpretation Rules

1. Prescription items are not unique patients.
2. NIC is not manufacturer revenue.
3. The QOF diabetes register is not the clinically eligible GLP-1 population.
4. DAUR and DADI are comparative analytical proxies, not treatment penetration measures.
5. Aggregate prescribing data cannot demonstrate patient-level switching.
6. High percentage growth from a small base should not be interpreted as equivalent to large commercial impact.
7. Low DADI does not automatically indicate commercial opportunity.
8. Geographic conclusions should be evaluated alongside disease burden, local pathways, eligibility and policy.
9. QOF prevalence, registered population and monthly prescribing data are not perfectly temporally aligned.
10. The analysis is intended to support commercial investigation, not clinical decision-making.